Nicholas Eisenberg, Ph.D.

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Experience

Senior Scientist - Machine Learning *June 2022 - Present*

Nevada National Security Site

- **LLM Inference Platform (Piccata):** Architected, deployed, and administer a multi-user LLM platform on Linux GPU servers, exposing internally hosted models through OpenAI-compatible `vLLM` APIs. Built a `FastAPI` gateway for authentication, routing, usage tracking, and persistence; developed `libserverctl` for user, API-key, certificate, and usage administration; and built a `Streamlit` portal for requesting access and provisioning API tokens and client certificates. Operate the platform with rootless `Podman`, `Nginx`, TLS/mTLS, and Linux networking.
- **LLM-Assisted Image Annotation and Retrieval (Photo-Pilot):** Designed and deployed a `Streamlit` application on Linux for LLM-assisted image annotation and retrieval. Integrated multimodal LLMs served through Piccata, persisted annotations and metadata in SQL, and built an interface for querying tagged imagery. Deployed with `Podman` and `Nginx`.
- **Python LLM Client Library (ChatLib):** Developed and maintain an internal Python library built on the OpenAI SDK, providing a simplified interface to internally hosted LLMs with conversation management, streaming, and model configuration.
- **Retrieval-Augmented Generation (RAG):** Designed and deployed a RAG system for internal Python/C++ codebases and technical documents. Built language-aware ingestion with Python `AST`, `Jedi`, and `Clang`, semantic embeddings, and HNSW vector retrieval backed by locally hosted LLMs.
- **Internal Scientific Software Monorepo:** Maintain a large internal monorepo supporting ML and scientific applications, including shared Python libraries, release workflows, automated testing, CI/CD, and Linux deployment infrastructure.

Technical Skills

Languages: Python, Bash, Lua, SQL, Cypher, LaTeX

Libraries / Frameworks: PyTorch, vLLM, FastAPI, Pandas, SQLAlchemy, Streamlit, Matplotlib, Plotly

Infrastructure / Tools: Linux, Podman, Docker, Nginx, Git, CI/CD, SLURM, AWS, Vim, Tmux

Open-Source Projects

iron.nvim - Sole Maintainer *May 2024 - Present*

github.com/Vigemus/iron.nvim

Maintain a cross-platform Neovim REPL plugin written in Lua; develop features, review contributions, manage releases, and resolve community issues.

Education

Auburn University - Ph.D. *June 2022*

Stochastic Partial Differential Equations

University of Nevada, Las Vegas - B.S./M.S. *December 2015 / January 2018*

Applied Mathematics

Publications

1: Eisenberg, N., Chen, L. *Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics.* Stoch PDE: Anal Comp (2022).

2: Eisenberg, N., Chen, L. *Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term.* J Theor Probab (2024).